

The Next Ravi Varma

Ravi Varma-inspired image generation with Stable Diffusion 1.5 + LoRA

Milestone review: Milestones 1-2 complete (~50% of the proposal)



Generated by our v2 LoRA

Can a diffusion model learn Ravi Varma's visual language?

- **Raja Ravi Varma (1848-1906):** fused European academic oil realism with Indian epics, attire and iconography.
- **Inspiration:** The Next Rembrandt (2016) - learn a painter's style from his corpus and create new works.
- **Our approach:** Stable Diffusion 1.5 stays frozen; a small LoRA adapter (3.2M parameters, 0.37% of the UNet) learns the style behind a trigger token <rvvarma>.
- **End goal (4 milestones):** epic scene text -> prompt expansion -> style LoRA + pose ControlNet -> evaluated image in a web panel.



Hamsa Damayanti - Raja Ravi Varma (training image)

Progress against the proposal



Milestone 1

Corpus curation & multimodal captioning

Complete



Milestone 2

Style-injected LoRA fine-tuning

Complete - trained & evaluated



Milestone 3

Text parsing & semantic scene generation

Not started



Milestone 4

ControlNet, evaluation framework, web panel

Not started

~50%

of the proposal delivered

2 of 4 milestones complete, with a trained model, generated images and a paired evaluation

What has been built: the end-to-end pipeline



- **Milestone 1:** first four stages (local machine, Apple M-series)
- **Milestone 2:** LoRA training on a Colab T4 GPU, generation + evaluation; weights load back on the Mac
- **Engineering:** YAML configs, reproducible CLI scripts, Colab notebook, 101 automated tests (99 pass, 2 run only without CLIP)

Curating 200 downloads into a clean training set

200

downloaded from Wikimedia Commons
(193 public domain, provenance kept)

134

accepted after validation
& curation

122 + 12

training images +
held-out evaluation set



66 excluded, each with a recorded reason

- **26 upscaled copies** - originals as small as 200 px (checked against Commons)
- **33 duplicates** - same artwork as prints/crops/re-scans; best copy kept
- **7 non-paintings** - stamps, gallery photos, panel grids, text labels



Excluded: postage stamp | gallery photo | multi-panel grid | 200 px original upsampled

Aspect-ratio buckets (e.g. 448x640) instead of square crops: heads and feet are no longer cut off.

Structured captions from a vision-language model



Training caption (real, from the manifest)

<rvvarma>, Hamsa Damayanti, a woman in a red sari feeding a white swan, wearing red dress, necklace jewelry, sitting, calm expression, in park, single-figure composition, vertical format, mid-tone lighting, warm red and ochre palette

- **BLIP-large** writes the scene sentence; **BLIP-VQA** answers a fixed template: people, clothing, colour, jewelry, pose, action, expression, objects, setting, day/night.
- **Grounded, not guessed:** follow-ups are gated ("what jewelry?" only if "wearing jewelry?" = yes); non-answers and anachronisms ("phone", "smoking") are filtered; all Q&A saved for audit.
- **Style left to the trigger token:** captions describe content, so <rvvarma> absorbs the painterly style; frames are named so they are not learned.
- **Fits CLIP's 77-token limit;** 134 images captioned in 29 min on a laptop.

LoRA fine-tuning on a Colab T4 GPU

1500

optimizer steps
(~49 epochs)

58 min

on one Tesla T4
(fp16)

3.2M

trainable params
(0.37% of UNet)

12.8 MB

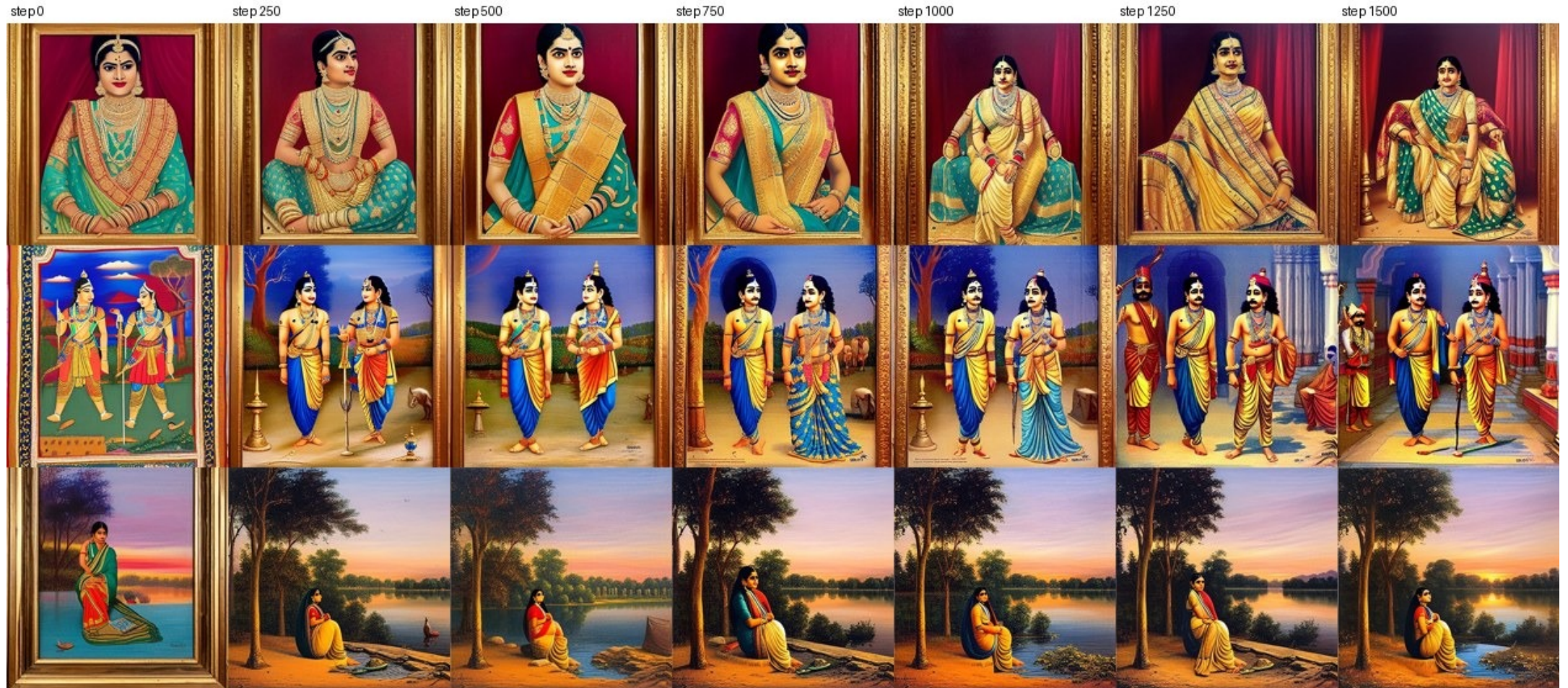
final LoRA
weights file

Setting	Value
Base model	Stable Diffusion 1.5 (frozen)
LoRA rank / alpha / dropout	16 / 16 / 0.05
Target modules	to_q, to_k, to_v, to_out.0
Learning rate	1e-4, 100 warm-up steps
Batch x grad. accumulation	1 x 4, seed 42

Fixed along the way

- LR warm-up was silently 4x too long - fixed
- Base-model repo removed from Hugging Face - moved to the official mirror
- Colab package conflicts (xformers, bitsandbytes) - removed, documented
- Every checkpoint now exports loadable weights; 20-step smoke test before the full run

The style emerges during training



Same 3 prompts and seeds at every checkpoint. Step 0 = untrained adapter. Frames disappear, a seated maharani pose and a pillared court appear after step 1000.

Base model vs our LoRA

BASE SD1.5



+ LoRA



"royal Indian woman ... palace interior"

BASE SD1.5



+ LoRA



"royal Indian portrait ... jewelry"

BASE SD1.5



+ LoRA



"woman seated beside a window ..."

- **Staging moves toward Varma:** full-length figures with pillars, carpets and lamp stands; maharaja-style court portraits; seated poses with draped saris.
- **No gilt frames:** 0 of 15 LoRA images vs 5 of 15 base images (same negative prompt).
- **Rendering:** warmer, darker painted backgrounds instead of glossy digital 'calendar art'.

What works, and what the LoRA gets wrong



Generated locally on a MacBook with the trained weights (seed 7)



Works

- Varma-like staging, palette and soft painted lighting on single figures
- Generalises to unseen prompts and seeds; no memorised copies (max similarity to any training image 0.88)
- Weights trained on Colab load and run on a laptop

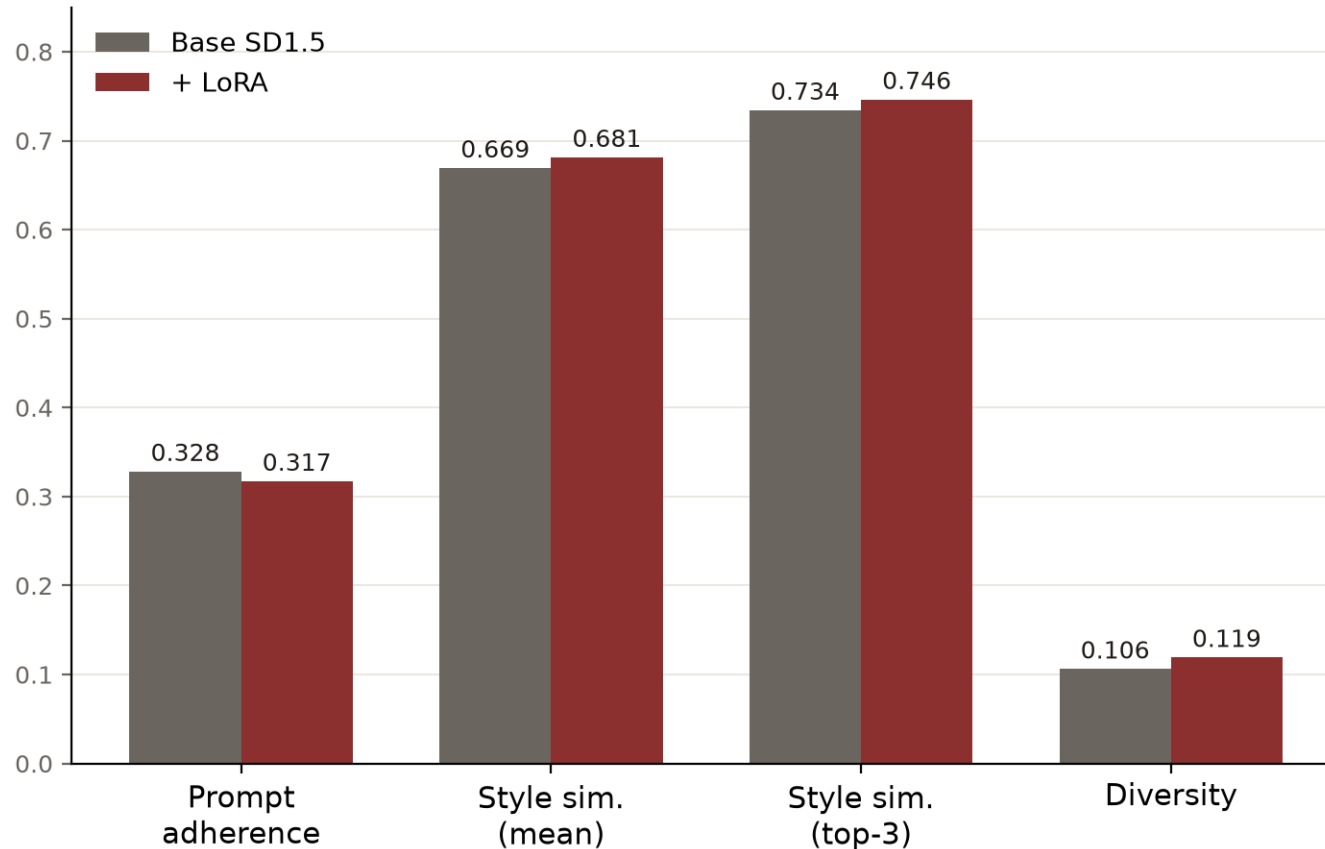


Problems found

- Moustache-like shadow on some women's upper lips (not in the base model)
- Signature-like scribbles in corners - learned from printed signatures on oleographs
- Crowded mythological scenes stay saturated 'poster art'; hands remain weak

Automatic metrics: a small, inconsistent shift

CLIP ViT-B/32 metrics: LoRA vs base (same prompts and seeds)



Base SD1.5 vs + LoRA: prompt adherence 0.328 vs 0.317; style similarity mean 0.669 vs 0.681, top-3 0.734 vs 0.746; diversity 0.106 vs 0.119.

- **Style similarity** to 12 held-out paintings: +0.012 (LoRA higher in 7 of 15 pairs).
- **Prompt adherence** -0.011, within noise for 15 pairs.
- **No memorisation**: no output is a near-copy of a training image.
- **Takeaway**: CLIP barely sees the composition and framing changes that are obvious to a viewer - it is a weak judge of painting style. A human rubric (quality, anatomy, style, prompt) is prepared for the next stage.

Next steps



Fix LoRA artifacts

Name or crop printed signatures in captions; find the portraits behind the upper-lip shadow; pick the best checkpoint with the human rubric.



Milestone 3: scene parser

LLM prompt expansion: epic narrative -> attribute-rich prompt with the trigger token (e.g. "Yudhishtira weeping over Karna at Kurukshetra").



Milestone 4: control & UI

OpenPose / Canny ControlNet for balanced, dignified poses; Gradio panel (text + pose -> image); fuller evaluation and checkpoint hub.

Summary

- A curated, provenance-tracked 134-image Ravi Varma corpus with grounded VLM captions
- A LoRA trained on a T4 that visibly moves SD1.5 toward Varma's staging and rendering
- Honest evaluation: clear visual gains, small CLIP gains, two artifacts to fix
- Code, data list, results and docs: github.com/f20240684-cmd/the-next-ravi-varma-v2



Thank you - questions?